# Make A Simple Power Supply For Adjustable Voltage And Current

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Sometimes a simple analogue power supply with adjustable output voltage and adjustable current-limiting function is needed. This article presents a simple power supply using an adjustable regulator LM350 that provides variable output up to 17V and maximum output current below 2A. LM350 has higher power dissipation as compared to commonly-available adjustable voltage regulator LM317 and, hence, has higher guaranteed output current. This power supply may be useful in laboratories and for hobby projects.

The circuit diagram of the power supply is shown in Fig. 1. It is built around bridge rectifier (BR1), adjustable voltage regulator LM350 (IC1), transistors BC327(T1) and BC337(T2), and a few other components.

Input to connector CON1 can be AC or DC. If you use an 18 to 20Vrms transformer with 2A current ratings, you can have output voltage $V_{OUT1}$ from 1.2V up to around 16.5V available at CON3, and $V_{OUT2}$ from 0V to 15V available at CON2. Input is protected with 2A fuse F1. Capacitors C3 and C5 (2200µF) are the main filtering capacitors.

Input voltage is limited by maximum input voltage of IC LM350. Maximum power dissipation of LM350 is around 25W.

According to the data sheet, input voltage of LM350 can be from around

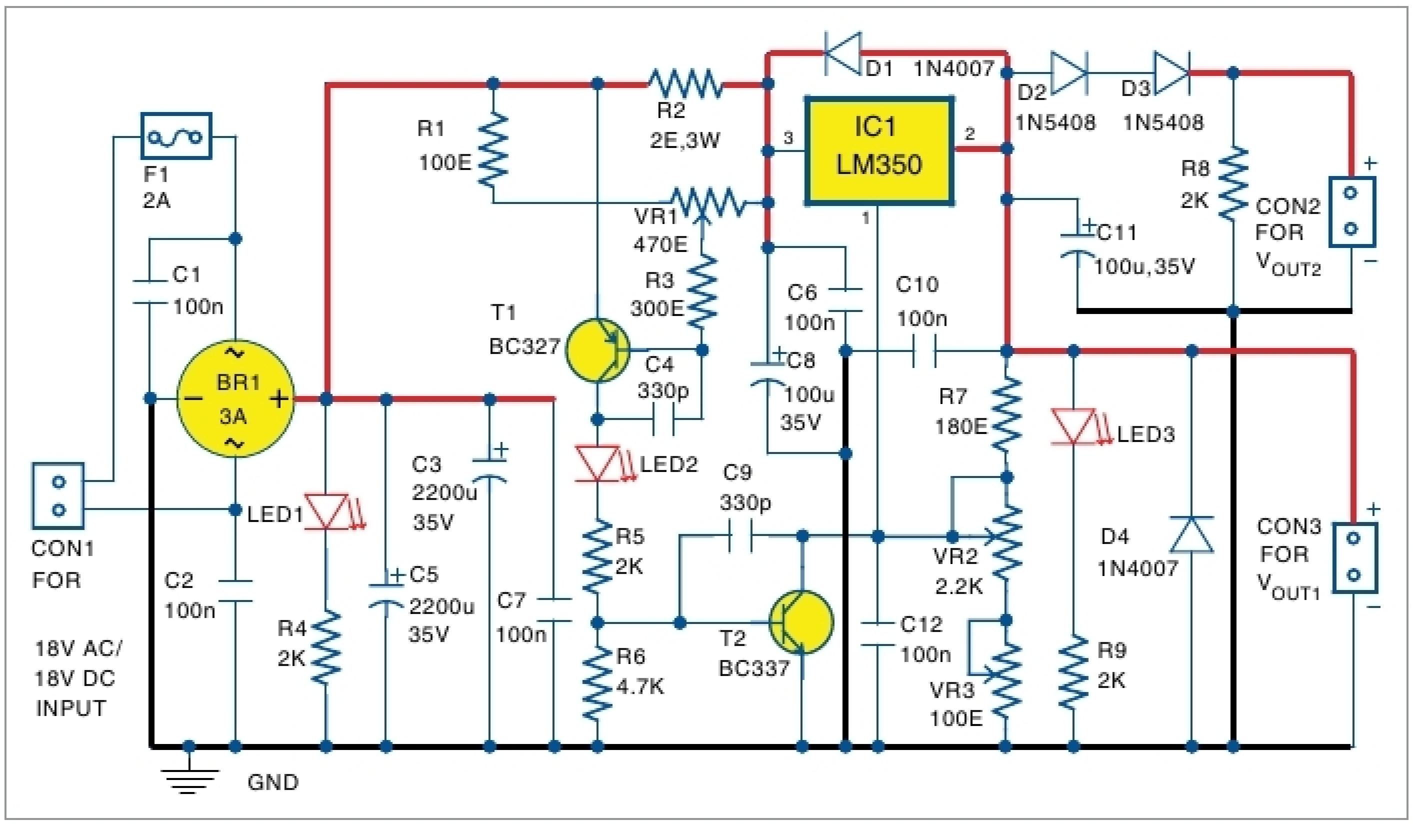


*Fig. 1: Circuit diagram of the simple power supply with adjustable voltage and current with LM350*

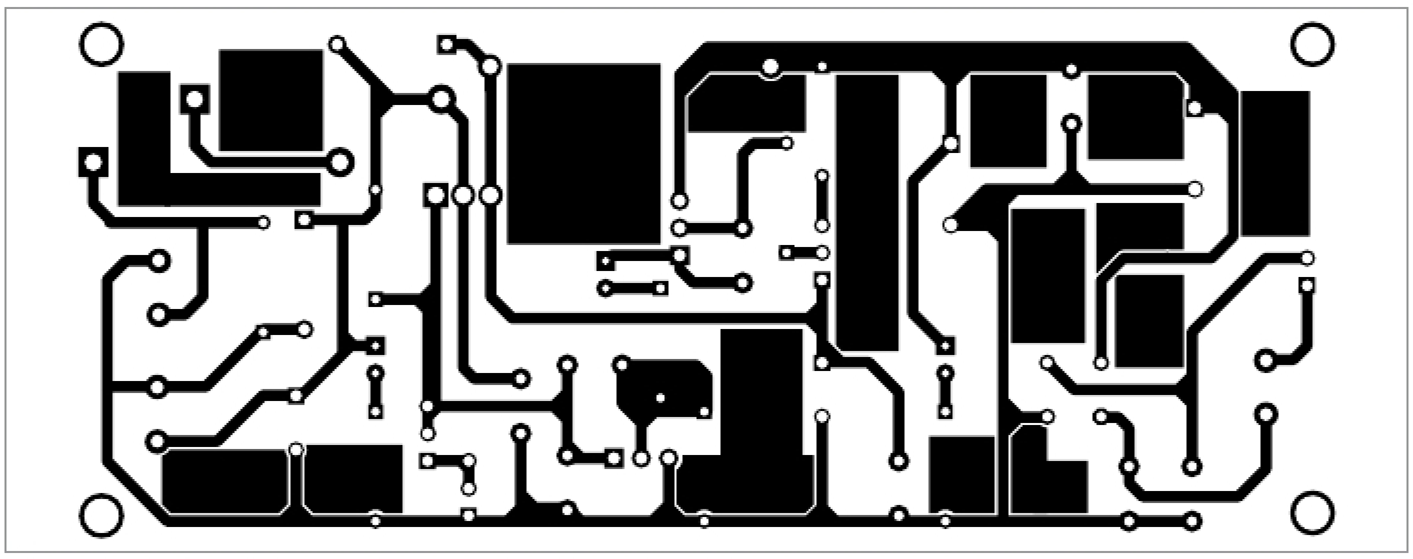

*Fig. 2: Actual-size PCB layout of the simple voltage adjustable power supply*

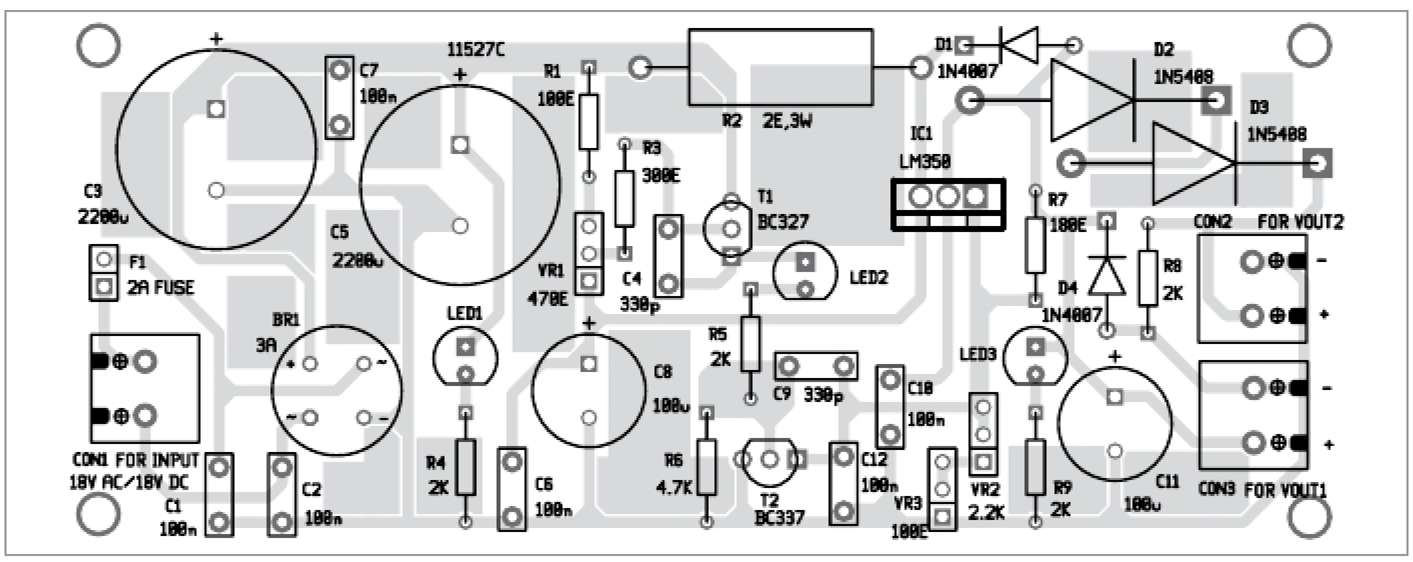

*Fig. 3: Components layout for the PCB*